

Analysis of a Bio-Inspired Myriapod Robot with a Modified Spine and Legs Linkage

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Abstract

This report details the design, modeling, and analysis of a bio-inspired terrestrial locomotion mechanism featuring two significant modifications to a baseline **myriapod** microrobot. Inspired by the work of Hoffman and Wood [1], our project investigates the combined dynamic effects of an asymmetrically compliant spine and a more biologically realistic leg mechanism. We propose a design that (1) replaces the anterior flexible Sarrus linkage of the spine with a rigid spine, creating a hybrid flexible-rigid backbone, and (2) replaces the original simple leg actuators with a foldable, single-actuator Jansen's linkage. The primary research objective is to analyze how this coupled modification of both spinal compliance and leg kinematics affects ambulatory behavior, gait stability, and overall system dynamics under a consistent control signal.

1 Introduction and Project Rationale

The field of bio-inspired robotics offers powerful paradigms for designing robust and efficient locomotion systems. The segmented, **myriapod-like** microrobot developed by Hoffman and Wood provides an excellent baseline, demonstrating the advantages of a flexible backbone. Our project extends this foundation by exploring a more complex and integrated engineering question: *How does the interplay between non-uniform spinal flexibility and a mechanically complex, bio-realistic leg linkage influence the performance of a multi-legged robotic system?*

By modifying both the spine and the legs, we move beyond studying isolated components to investigating a more holistic system redesign. The original robot uses simple rotational leg motions, whereas a Jansen's linkage produces a more sophisticated foot trajectory from a single rotary input. Our central hypothesis is that the dynamic coupling between this leg mechanism and an asymmetrically stiffened spine will yield unique locomotor characteristics not present in the original, fully flexible, simply-actuated design.

1.1 Project Scope

To ensure the project is feasible while addressing our core research question, we will constrain our investigation as follows:

- **Dual System Modification:** Our primary investigation involves the parallel implementation of two key changes:
 1. **Hybrid Spine:** The anterior flexible Sarrus linkage (between segments one and two) will be modeled as a rigid connector. The posterior linkage (between segments two and three) will remain flexible.

2. **Jansen’s Linkage Legs:** The original two-actuator leg mechanisms will be replaced with a one-degree-of-freedom (1-DOF) Jansen’s linkage, designed to be manufacturable using foldable techniques.

- **Three-Segment Model:** We will model a three-segment version of the robot, which is the minimum required to implement a stable alternating tripod gait.
- **Control Signal Parity:** To isolate the effects of the mechanical modifications, the control input (rotary drive signal for the Jansen’s linkage crank) will be kept consistent across all simulations. This ensures a fair comparison of the resulting system dynamics.

1.2 Impact

This project is significant because it explores the systemic effects of combining different bio-inspired design principles. A robot with a rigid anterior section may gain straight-line stability, while the Jansen’s linkage could offer improved ground clearance and efficiency. The results could inform the design of future foldable robots that require a careful balance of stability, efficiency, and maneuverability, demonstrating how localized stiffness and complex leg kinematics can be synergistically employed.

1.3 Topic Fit: Foldable Robotics

The original microrobot is fundamentally based on the Smart Composite Microstructures (SCM) process. Our proposed modifications fully adhere to these principles. Both the new rigid spine component and the multi-bar Jansen’s linkage will be designed as 2D patterns that can be fabricated and folded into complex 3D structures. This project thus showcases the versatility of foldable manufacturing in creating intricate mechanisms with localized stiffness variations and complex kinematic functions.

2 Background Research

Our design is a direct evolution of the myriapod-like microrobot presented by Hoffman and Wood. A thorough review of this and other relevant literature is crucial for grounding our work.

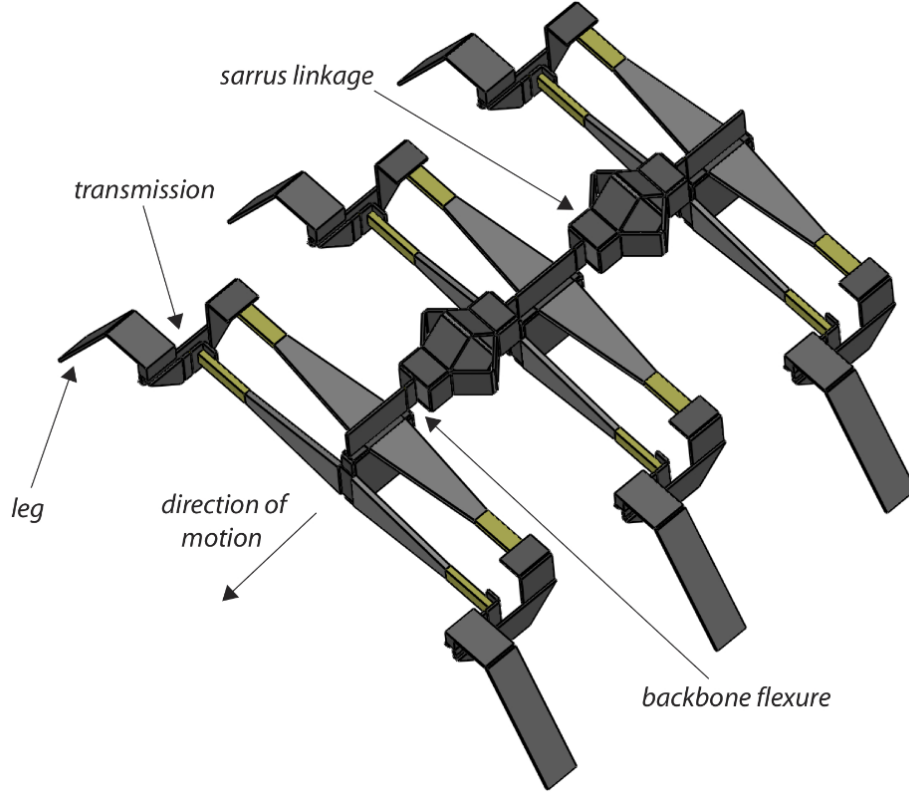


Figure 1: The original microrobot design serving as the baseline for our modifications.

2.1 Primary Inspiration

The paper "Myriapod-like ambulation of a segmented microrobot" by Hoffman and Wood [1] provides the complete blueprint for our project's baseline. It details the SCM-based design, a comprehensive Euler-Lagrange dynamic model, and fabrication details. Our work will directly adapt their dynamic model to account for both the asymmetrical rigidity of our modified backbone and the new kinematics of the Jansen's linkage legs.

2.2 Supporting Literature

To ground our work, we surveyed literature covering compliant spines, foldable mechanisms, and integrated system dynamics.

First, research into spinal compliance provides a strong motivation for our hybrid backbone design. The work by Bing et al. [3] on a bio-inspired mouse robot demonstrates experimentally that lateral spinal flexion can significantly improve motor performance. This supports our hypothesis that modifying spinal properties will have a substantial impact on robot dynamics, which we extend by exploring *asymmetrical* compliance to study how localized stiffness affects stability.

Second, our use of a foldable Jansen's linkage is informed by established research in folded robotics. The prototyping process for folded millirobots detailed by Hoover and Fearing [6] and the design of a foldable hexapod by Önal [7] are highly relevant. Together, they demonstrate the feasibility of fabricating complex, multi-legged systems from planar laminates, providing a clear precedent for manufacturing our proposed leg mechanism within the SCM paradigm.

Finally, our holistic approach is validated by literature that examines the interplay between robotic subsystems. The work by Eckert et al. [9] directly compares different spine and leg designs, supporting our premise that these components form a tightly coupled system. Furthermore, the

review by Dürr et al. [8] on autonomous hexapedal locomotion provides a biomimetic foundation for our use of the alternating tripod gait, grounding our dynamic analysis in established principles of stable multi-legged movement. These papers collectively justify our investigation into the combined effects of a hybrid spine and complex leg kinematics.

2.3 Project Novelty

The novelty of our project lies in the holistic investigation of a coupled spine-leg modification. While previous work has often focused on either spinal compliance or leg mechanism design in isolation, our research explores the intersection of these two domains. By introducing a hybrid spine and a Jansen’s linkage simultaneously, we can study how localized stiffness interacts with a more complex, bio-realistic foot trajectory, providing new insights into the integrated design of ambulatory robots.

3 System Specifications and Performance Metrics

The initial physical parameters for our model are derived directly from Table 1 in Hoffman and Wood [1], ensuring a consistent baseline for comparison of segment mass and inertia.

Table 1: Key Physical Parameters for a Single Robot Segment (from [1])

Parameter	Symbol	Value
Segment Mass	m	250 mg
Body Half-Length	L_b	3 mm
Body Half-Width	w_b	10 mm
Segment Inertia	I_{cm}	$9.3 \times 10^{-9} \text{ kg}\cdot\text{m}^2$
Max Actuator Torque (Original)	τ_{max}	$41.4 \text{ } \mu\text{Nm}$

3.1 Initial Calculations

To establish goal performance metrics, we can perform initial estimates. The force required to accelerate a single segment is a useful baseline. Assuming a desired forward acceleration (a) of 0.1 m/s^2 :

$$F_{net} = ma = (250 \times 10^{-6} \text{ kg}) \times (0.1 \text{ m/s}^2) = 25 \text{ } \mu\text{N}$$

This net force must be generated at the foot during the stance phase. Using the Jacobian of the Jansen’s linkage, we can translate this required foot force back to a required input torque at the driving crank, which will inform our actuator selection and design.

4 Mechanism Design

4.1 Analogous Mechanism (Physical Prototype)

A low-fidelity physical prototype will be constructed from paper or cardboard to validate the geometry and range of motion of our modified design. This model will integrate both the hybrid spine (one rigid, one flexible connection) and the foldable Jansen’s linkage legs. This will allow us to physically verify the leg’s foot path and observe the interaction between segments before committing to a digital model.

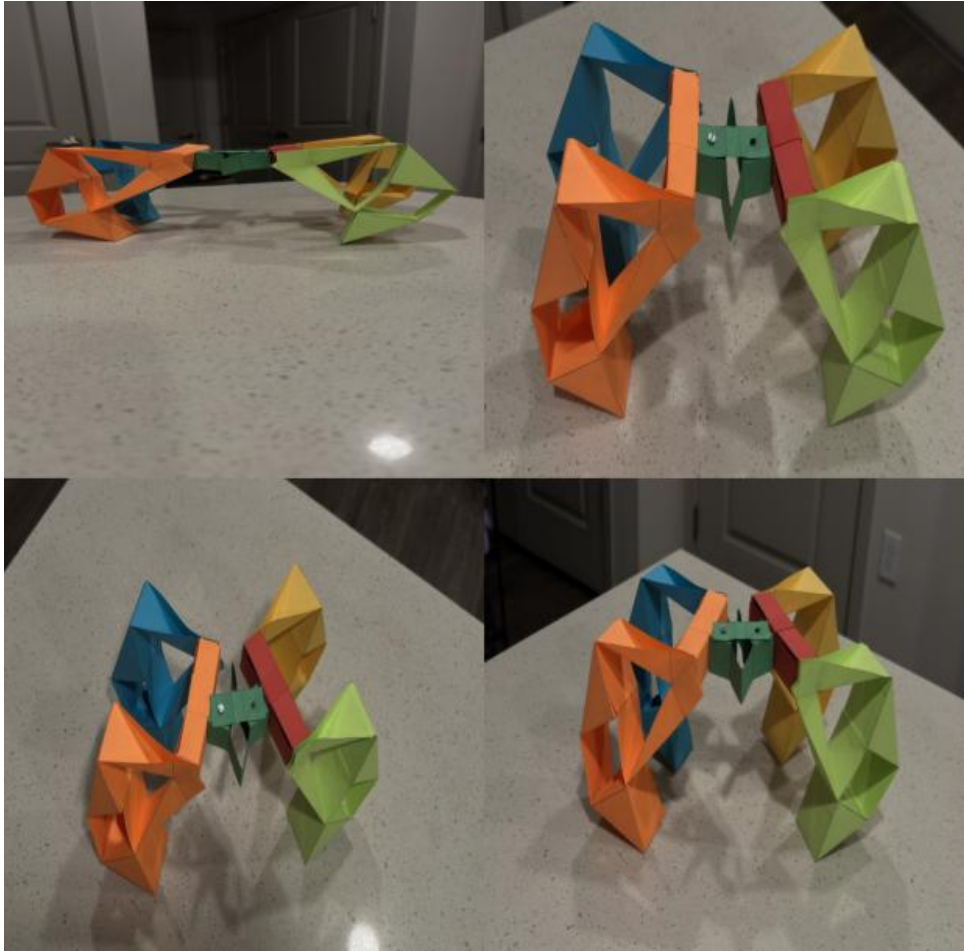


Figure 2: Physical prototype of the proposed mechanism.

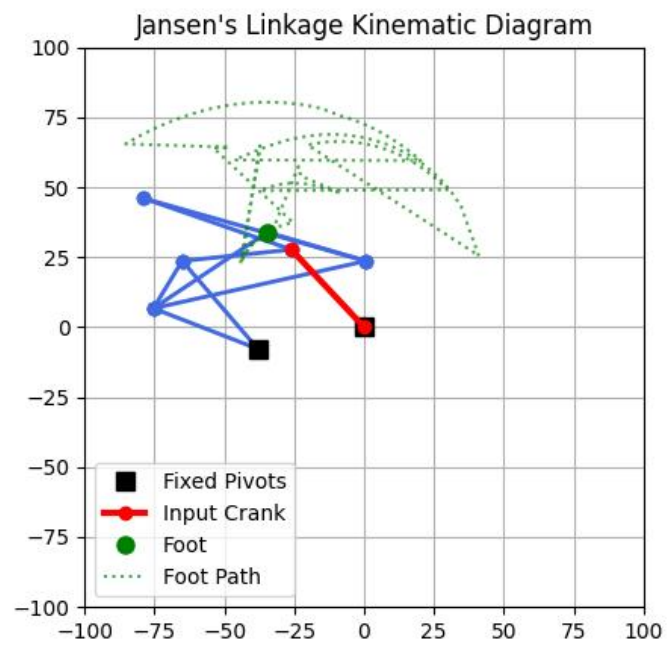


Figure 3: Kinematic diagram of the proposed leg mechanism.

4.2 Proposed Mechanism (Digital Model)

A detailed digital model of the proposed mechanism will be created using vector drawing software. This model will explicitly show the three-segment chassis with the hybrid spine and one Jansen’s linkage leg, fully labeled for kinematic analysis. Labels will include reference frames, joint angles (e.g., crank angle ϕ_i), and all link lengths (l_i).

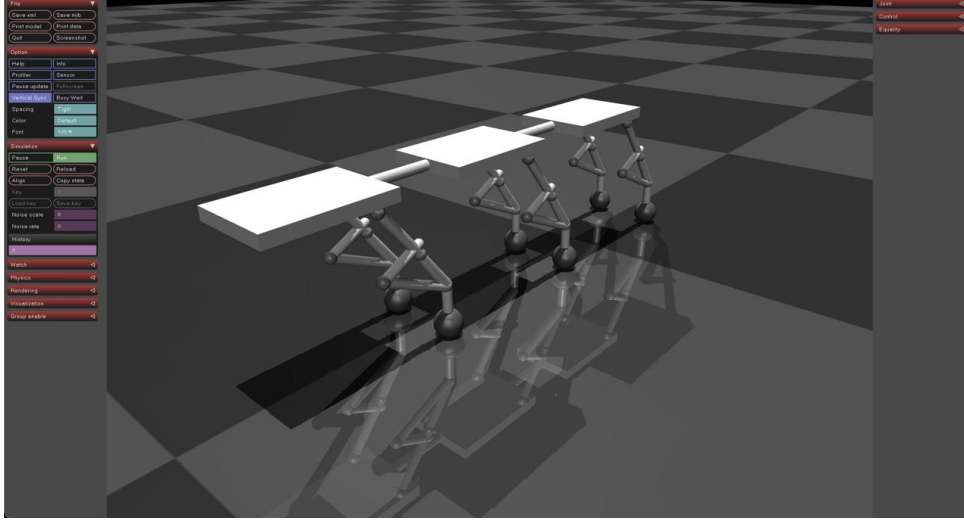


Figure 4: Labeled digital model of the proposed mechanism with kinematic variables and reference frames clearly labeled for analysis.

5 Kinematic and Dynamic Modeling

5.1 Kinematic Model

A kinematic model of the mechanism will be developed in Python. This model will be hierarchical:

1. **Leg Kinematics:** First, we will model the 1-DOF Jansen’s linkage to define the foot position as a function of the input crank angle, ϕ_i .
2. **System Kinematics:** This leg model will then be integrated into the three-segment chassis model. The position and orientation of each segment will be defined based on the state of the six leg actuators and the constraints imposed by the spine.

From this model, we will numerically derive the system Jacobian, which relates the angular velocities of the leg cranks to the velocities of each segment’s center of mass and the forces at the feet.

5.2 Dynamic Analysis

The dynamic model will be adapted from the Euler-Lagrange formulation in [1]. Our modifications will be twofold:

1. The kinetic and potential energy terms will be reformulated based on the more complex kinematics of the Jansen’s linkage.
2. The potential energy calculation for the backbone will be altered: the spring constant terms (k_l and k_t) will be set to effectively infinite (or removed) for the anterior connection to model it as rigid, while they will be retained for the flexible posterior connection.

The full equations of motion will be solved numerically to simulate the robot’s movement and compare it to the baseline model.

6 Discussion

6.1 Answers to Discussion Questions

1. **Degrees of Freedom (DOF) and Motors:** The complete three-segment robot has six legs. Each Jansen’s linkage leg is a 1-DOF mechanism driven by a single motor. Therefore, the entire system has 6 actuated DOF. In our planar dynamic model, we consider the rotation of each of the three segments’ bodies (θ_i) and the input crank angle for one leg per segment (ϕ_i). This results in 6 total generalized coordinates. With only one actuator per segment controlling the leg, the body rotation is a passive degree of freedom, making the system underactuated.
2. **Determining Remaining DOF:** The state of the unactuated DOF (the body rotation, θ_i) is not directly controlled. Its motion is an emergent property of the system’s dynamics—specifically, the interplay between the actuated leg motion (via the Jansen’s linkage), the inertial properties of the segment, and the constraint forces from the ground and the hybrid spine. The dynamic simulation is the tool used to determine the behavior of these passive degrees of freedom.
3. **Estimating End-Effector Forces and Speeds:** The expected end-effector (foot) forces are the ground reaction forces required for propulsion. These can be estimated by calculating the required actuator torque at the crank and transmitting it through the linkage using the Jacobian of the Jansen’s mechanism ($F_{foot} = J^{-T} \tau_{crank}$). Similarly, end-effector speeds are calculated from the actuator’s angular velocity using the forward kinematics and Jacobian ($v_{foot} = J \dot{\phi}_{crank}$) to ensure the desired step length and cadence are achieved.

7 Implementation Plan and Experimental Procedure

To systematically achieve our research goals, the following implementation plan will be followed:

1. **Baseline Model Replication:** First, develop a Python simulation that replicates the dynamic model and results of the fully flexible, simple-legged robot from Hoffman and Wood [1]. This validates our simulation environment and provides a verified baseline.
2. **Jansen’s Linkage Leg Model:** Develop and validate a standalone kinematic and dynamic model of our proposed foldable Jansen’s linkage. Determine its foot trajectory and force/torque transmission characteristics.
3. **Integrated Model 1 (Jansen Legs, Flexible Spine):** Integrate the validated Jansen’s linkage model into the baseline three-segment chassis, initially retaining the fully flexible spine. Simulate this system to isolate the effect of the leg modification.
4. **Integrated Model 2 (Jansen Legs, Hybrid Spine):** Modify the model from the previous step to incorporate the hybrid spine by rigidifying the anterior connection in the Euler-Lagrange equations. This represents our final proposed design.
5. **Comparative Simulation and Analysis:** Run simulations on all three models (baseline, integrated 1, integrated 2) using identical control signals. Perform a comparative analysis of key performance metrics, including:
 - **Forward Velocity and Efficiency:** Quantify locomotion speed and energetic cost.
 - **Gait Stability:** Analyze the trajectory of the center of mass and inter-segment angles.
 - **Robustness:** Potentially test performance on slightly uneven terrain in simulation.

6. **Physical Prototyping and Final Reporting:** In parallel, construct the low-fidelity cardboard analogous mechanism to visualize and refine the kinematics. All findings, simulations, models, and discussions will be compiled into the final report and presentation, ensuring all assignment deliverables are met.

8 Conclusion

This report outlines a comprehensive plan to design, model, and analyze a bio-inspired robotic mechanism with significant modifications to both its spine and leg structure. By integrating a hybrid rigid-flexible spine with a complex Jansen's linkage leg, we aim to provide quantitative insights into the coupled effects of these design choices on terrestrial locomotion. This work will contribute to a deeper understanding of how to synergistically combine different bio-inspired principles to create small-scale ambulatory robots with enhanced stability and performance.

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